MATH221: MATH IV (Probability and Statistics)

COURSE INFORMATION			
Course Code:	MATH 221	Lecture Contact Hours:	42
Course Title:	MATH IV (Probability and Statistics)	Credit Hours:	3.0

PREREQUISITE

Course Code: MATH 221

Course Title: MATH IV (Probability and Statistics)

CURRICULUM STRUCTURE

Outcome-Based Education (OBE)

RATIONALE

This course is one of the fundamental courses of Computer Science and Engineering that is needed to understand elementary probability theory and practical data analysis techniques and students can implement a small report summarizing, presenting and analyzing an appropriate data set.

OBJECTIVE

Students will be able to-

1. Achieve a sound understanding of the theoretical and practical knowledge of statistics.
2. Impart with fundamental knowledge about descriptive statistics and their applications.
3. Understand probability as a model for uncertainty.
4. Understand the basic probability axioms and the moments of discrete and continuous random variables as well as be familiar with probability distribution.

LEARNING OUTCOMES & GENERIC SKILLS

No.	Course Learning Outcomes (Upon completion of the course, the students will be able to)	Bloom's Taxonomy	CP	CA	KP	Assessment Methods

CO1	**Understand** the terminologies of basic probability and statistical theory with data analysis techniques and interpret the behavior of various probability distributions.	C2	1	-	2	Class Test, Mid, Assignment, Final Exam
CO2	**Apply** the statistics for testing the significance of the given large and small sample data by using different tests and use random variable to solve suitable random phenomena.	C3	1	-	2	Class Test, Mid, Assignment Final Exam

C: Cognitive; P: Psychomotor; A: Affective; S: Soft-skills (CT: Critical Thinking, TW: Teamwork)

COURSE CONTENT

Introduction: Statistics; Types of statistics; Population and sample; Parameter and statistic; Types of Variable; Characteristics, scope and applications of statistics ; Limitations and misuses of statistics.

Collection of Data: Meaning of data, types of data; Sources of statistical data; Data collection tools; Level of measurement.

Presentation of Data: Constructing frequency distribution and relative frequency distribution: Qualitative and quantitative data; Cumulative
Frequency distribution; Graphic presentation of a Frequency distribution with merits and demerits.

Measures of Central Tendency and Location: Arithmetic Mean, Geometric Mean, Harmonic Mean, Weighted Mean, Median and Mode with uses, advantages, and limitations.

Measures of Dispersion: Meaning of dispersion; measures of dispersion; absolute measures of dispersion, relative measures of dispersion; Application of different measures of dispersion; Skewness, Kurtosis and their measures.

Correlation and Regression: Bivariate data, scatter diagram, simple correlation, rank correlation, simple and multiple regression

Test of Hypothesis: Hypothesis, Mean test, Several Mean test

Probability: Sample Space, Tree diagram, Probability, Marginal probability, Joint probability, Conditional probability, Probability distribution.

SKILL MAPPING

No.	Course Learning Outcome	PROGRAM OUTCOMES (PO)											
		1	2	3	4	5	6	7	8	9	10	11	12
CO1	**Understand** the terminologies of basic probability and statistical theory with data analysis techniques and interpret the behavior of various probability distributions.	H											
CO2	**Apply** the statistics for testing the significance of the given large and small sample data by using different tests and use random variable to solve suitable random phenomena.	H											

(H - High, M - Medium, L - Low)

JUSTIFICATION FOR CO-PO MAPPING

Mapping	Level	Justifications
CO1-PO1	High	Understand the appropriate techniques for data collecting and analysis process and compute the possibility of various probability distributions.
CO2-PO1	High	Applying different statistical testing procedure for hypothesis testing in large and small range of sample distributions and finding expected results for different real life situation.

TEACHING LEARNING STRATEGY

Teaching and Learning Activities	Engagement (Hours)
Face-to-Face Learning	
Lecture	42
Practical / Tutorial / Studio	-

Student-Centered Learning	-
Self-Directed Learning Non-face-to-face learning Revision Assessment Preparations	 42 21 21
Formal Assessment Continuous Assessment Final Examination	 2 3
Total	131

COURSE SCHEDULE

Class	Topics/Assignment	COs	Reading Reference	Lecture Outcomes/ Activities
01	Discussion of Course Details and why we study probability and statistics?	1	2	Lecture Discussion
02	Basic concept of Probability, Sets, Sample space, Event.	1	1 & 2	Lecture, Inquiry- Guided Instruction and solving problem, Feedback.
03	Meaning and Definition of Statistics; Types of statistics; Population, sample; Parameter and statistic, Variable and types of variable; Characteristics, limitations and misuses of statistics.	1	2	Lecture Discussion, Verbal feedback
04	Conditional probability and Tree diagram concept	1	2	Lecture, Student learning by doing.
05	Conditional probability and Tree diagram concept	1	2	Lecture, Problem discussion, Feedback.
06	Meaning and types of data, Sources of statistical data; Data collection tools; and Constructing frequency distribution and relative frequency distribution: Qualitative and quantitative data;	1	1	Lecture, Student learning by discussion
07	Probability distribution	1	2	Basic discussion on Probability.
08	Probability distribution	1	2	Problem discussion and engage students

09	Graphical representation (Polygon, Pie-chart, Histogram etc.) of a Frequency distribution with merits and demerits.	1	2	Lecture (discussion on graph Plotting)
10	Marginal Probability	2	2	Lecture, Student Interaction
11	Class Test-01: (W-1-3)	1	2	Student feedback on CT question
12	Average or measures of central tendency, Arithmetic Mean, Properties of Arithmetic Mean, Geometric Mean and their problems with uses, advantages, and limitations.	1	2	Lecture, Inquiry- Guided Instruction and solving problem, Feedback.
13	Marginal Probability	2	2	Real life based problem discussion and Homework.
14	Joint Probability	2	2	Lecture, Student Interaction
15	Harmonic Mean, Quadratic mean (Root mean square), Weighted Mean, and their problems with uses, advantages, and limitations.	1	2	Lecture, Solving problem step by step, Q/A.
16	Joint Probability	2	2	Real life based problem discussion and Homework.
17	Assignment based on the class lecture: 1-16 and previous class discussion	2	1 & 2	----------
18	Class Test-02: (W 1 - 5)	1	1	Student feedback on CT question and solving problem
19	Independent and dependent probability with concept of marginal and Joint probability	2	2	Basic discussion and show their correlation
20	Independent and dependent probability with concept of marginal and Joint probability	2	2	Related problem discussion and student interaction
21	Harmonic Mean, Quadratic mean (Root mean square), Weighted Mean, and their problems with uses, advantages, and limitations.	1	1	Lecture, Student learning by doing.
	MID-TERM EXAMINATION			
22	Binomial Distribution	2	2	Basic discussion and recall previous class lecture

23	Median and Mode for group and ungroup data with uses, advantages, and limitations.	1	1	Discuss previous Lecture and co-relate with present concept
24	Measure of location (Quartile, deciles and percentiles) with uses, advantages, and limitations.	1	1	Lecture, Solving problem step by step, Q/A and taking verbal feedback.
25	Binomial Distribution	2	2	Solving practical problem using Binomial
26	The semi-interquartile range, the 10–90 percentile range and their problems.	1	1	Lecture, Q/A, and overview all previous class
27	Measures of dispersion (Deviation, Standard deviation, Variance, Coefficient of variance	1	2	Lecture Discussion, Problem constructing and solve
28	Poisson Distribution	2	2	Basic discussion and recall previous class lecture
29	Measures of dispersion (Deviation, Standard deviation, Variance, Coefficient of variance	1	1 & 2	Student engagement and solve their relevant problems
30	Skewness, Kurtosis and their measures.	2	2	Define the formula and basic discussion.
31	Poisson Distribution	2	2	Solving practical problem using Poisson
32	Apply Moment on Skewness, Kurtosis and their measures.	2	1 & 2	Solving problem step by step.
33	Class Test 03: (W: 8-10) and discuss about assigned Assignment (Statistics)	1	1 & 2	---------
34	Discussion on different Probability distribution	1	2	Engage student to understand their problem
35	Regression and Correlation	2	2	Basic explanation on topic with visualization
36	Regression and Correlation	2	2	Lecture, solving problem, Feedback.
37	Class Test 04: (W-8-10)	2	2	---------
38	Test of Hypothesis (X^2-test)	2	2	Lecture, Basic discussion and solving problem

39	Test of Hypothesis (T-test)	2	2	Lecture, Basic discussion and solving problem
40	Review of whole syllabus	2	1 & 2	Get student feedback about course
41	Class Test 04: Probability (Z-test)	2	2	Lecture, Basic discussion and solving problem
42	Review the whole syllabus	1,2	1&2	Review course and get student feedback
	TERM FINAL EXAMINATION			

ASSESSMENT STRATEGY

			CO	Blooms Taxonomy
Components		Grading		
Continuous Assessment (20%)	Test 1-4	10%	CO1 CO2	C2 C3
	Class Participation	5%	CO1	C2
	Assignment	5%	CO2	C3
Midterm		30%	CO1 CO2	C2 C3
Final Exam		50%	CO1, CO2	C2 C3
Total Marks		100%		

(CO = Course Outcome, C = Cognitive Domain, P = Psychomotor Domain, A = Affective Domain)

REFERENCE BOOKS

Text Books:

1. Murray R. Spiegel, John J. Schiller, R. Alu Srinivasan, Probability and Statistics
2. M. Nurul Islam, An Introduction to Statistics and Probability, 4th edition

Referance Books

1. Milan Holický, Introduction to Probability and Statistics for Engineers, Springer Publisher.
2. Matthew A. Carlton, Probability with Applications in Engineering Science and Technology
3. S.P. Gupta, M.P. Gupta, Business Statistics.

Course Teacher

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Signature of Course Teacher	Signature of HoD	Signature of Dean